Production Functions, Productivity, and Markups

C.Conlon

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Grad IO

Overview

- Why we estimate production functions: productivity facts, and two uses of $f(\cdot)$
- The identification problem: simultaneity, selection, and prices (TFPR vs TFPQ; Foster–Haltiwanger–Syverson)
- One equation, three families of estimators
  - dynamic panel (Blundell–Bond)
  - proxy / control function: Olley–Pakes, Levinsohn–Petrin, Ackerberg–Caves–Frazer
  - first-order conditions for gross output: Gandhi–Navarro–Rivers; plus Leontief in materials
- Endogenous productivity: exporting and R&D belong in the law of motion
- Markups from production data (De Loecker–Warzynski; multi-product firms), and where it breaks
  - revenue data; invertibility; market power; heterogeneous technology; input market power
- Aggregation (which costs, which average) and misallocation

Notation: producer j, period t. Lowercase letters are logs of uppercase levels, except prices: output price p_{jt} and input prices w_{jt}^X are levels, as in the rest of the course.

Why Production Functions?

Two Uses of a Production Function

$$y_{jt} = f(\mathbf{x}_{jt}) + \omega_{jt} + \varepsilon_{jt}, \qquad \mathbf{x}_{jt} = (k_{jt}, l_{jt}, m_{jt})$$

1. Productivity ω_{jt}: what is left once inputs are accounted for.
   - dispersion and persistence; reallocation and entry/exit
   - effects of deregulation, trade liberalization, exporting, R&D, management
2. Output elasticities $\theta^X_{jt} = \partial f / \partial x_{jt}$.
   - returns to scale, factor-biased technology
   - markups: $\mu_{jt} = \theta^X_{jt} / \alpha^X_{jt}$ with α^X the revenue share of a flexible input (De Loecker–Warzynski 2012)

Questions this lecture can answer:

- How much of aggregate productivity growth comes from reallocation and turnover?
- What can production data alone say about market power, and about its trend?

Facts About Producer-Level Productivity

- Dispersion: the 90–10 TFP ratio is about 1.9:1 within US 4-digit manufacturing (log gap 0.65, Syverson 2004), about 2:1 within 6-digit industries in the US and Canada, and above 5:1 (revenue-based) in China and India (Hsieh–Klenow 2009).
- It is not missing inputs: adding R&D stocks or detailed worker quality barely moves it (Fox–Smeets 2011: 90–10 ratio 3.74 $\longrightarrow$ 3.36).
- Persistence: annual AR(1) coefficients of 0.6–0.8; 0.3–0.4 with producer fixed effects.
- Correlates: more productive producers are larger, more profitable, grow faster, are more likely to survive, pay higher wages, and (for homogeneous goods) charge lower prices.

> *"I found the spectacle of economic models yielding large residuals rather uncomfortable, even when the issue was fudged by renaming them technical change and claiming credit for their 'measurement.'"*
> *— Zvi Griliches*

Sources: Syverson (2011, JEL); De Loecker and Syverson (2021, Handbook of IO).

What Is Productivity?

Hicks-neutral productivity Ω: $\quad Y = \Omega\, F(K, L, M).$

- Three equivalent views (De Loecker–Syverson 2021):
  - a shifter of the isoquants;
  - an output-to-input ratio, $\Omega = Y/F(\cdot)$: total factor productivity (TFP);
  - a shifter of the cost curve: higher Ω, lower cost at every Y.
- Labor productivity Y/L moves with K/L even when technology does not, so we usually want TFP. TFP needs a functional form and a measure of capital.
- Data usually record revenue, not quantity. In logs:

$$\text{TFPR}_{jt} = \text{TFPQ}_{jt} + \ln p_{jt}.$$

- TFPR is not the residual of a production function estimated with revenue on the left. We derive the gap below, after Klette–Griliches.

What Moves Productivity? (Syverson 2011)

Inside the producer

- Management practices: the interquartile range of management scores is worth 3.2–7.5% of TFP (Bloom–Van Reenen 2007).
- Input quality: worker skill, capital vintage.
- IT and R&D (Doraszelski–Jaumandreu 2013: R&D also makes productivity riskier).
- Learning by doing: Lockheed L-1011 hours per plane halved by the 30th plane (Benkard 2000).
- Product innovation; firm structure.

Outside the producer

- Spillovers (Liberty ships).
- Competition: US iron ore labor productivity doubled 1980–85 under import pressure (Schmitz 2005).
- Deregulation: telecom equipment (Olley–Pakes 1996).
- Trade; flexible input markets.
- Misallocation across producers (Hsieh–Klenow 2009; later).

All of these need a credible estimate of ω_{jt}, which needs a credible estimate of f. Firm choices among them (R&D, exporting) belong in the law of motion for ω (later).

The Identification Problem

Setup and Timing

$$y_{jt} = f(k_{jt}, l_{jt}, m_{jt}) + \omega_{jt} + \varepsilon_{jt}, \qquad \omega_{jt} = g(\omega_{jt-1}) + \xi_{jt}, \qquad \mathbb{E}[\xi_{jt} \mid \mathcal{J}_{jt-1}] = 0.$$

- ω_{jt} is known to the firm when it chooses period-t inputs ($\omega_{jt} \in \mathcal{J}_{jt}$); ε_{jt} is not (an ex post shock or measurement error).
- First-order Markov productivity. In this lecture ξ_{jt} is the productivity innovation, not BLP's demand unobservable; demand shifters will be written δ_{jt}. With an AR(1), $g(\omega) = \rho\,\omega$ (ρ = persistence, not the nesting parameter).
- Input timing (the GNR taxonomy):
  - predetermined: $k_{jt} \in \mathcal{J}_{jt-1}$, chosen through investment i_{jt-1};
  - flexible: chosen at t after seeing ω_{jt}, no dynamic implications (materials; often labor);
  - dynamic: lagged choices affect current ones (adjustment costs).
- Data: many producers, few periods ($N \to \infty$, T fixed).

Simultaneity (Marschak and Andrews 1944)

Regress output on inputs and the slope is θ^X plus a transmission bias, $\partial \mathbb{E}[\omega_{jt} \mid \mathbf{x}_{jt}]/\partial x$. To see its size, take Cobb–Douglas value added with k predetermined, l flexible, price taking, and a firm-level price ratio $d_{jt} \equiv \ln(w^L_{jt}/p_{jt})$ independent of ω. The labor FOC gives

$$l_{jt} = \frac{\ln \beta_l - d_{jt} + \beta_k k_{jt} + \omega_{jt}}{1-\beta_l} \quad \Longrightarrow \quad \operatorname{plim} \widehat{\beta}_l^{OLS} = \beta_l + (1-\beta_l)\,\frac{\sigma^2_\omega}{\sigma^2_\omega + \sigma^2_d},$$

with variances net of k.

- With no price dispersion ($\sigma^2_d = 0$), ω is an exact function of (l, k) and $y_{jt} = l_{jt} + \text{const} + \varepsilon_{jt}$: OLS puts 1 on labor and 0 on capital, whatever the truth.
- The more an input responds to ω rather than to prices, the closer its coefficient gets to 1. In data, OLS overstates the materials elasticity by 34% (Chile and Colombia, GNR 2020) and labor in Olley–Pakes's data. Biases on less flexible inputs can go either way.
- σ^2_d is the variation an input-price IV would use, but firm-level prices are rarely observed and often reflect quality or an upward-sloping supply curve (Griliches–Mairesse 1998).
- $\sigma^2_d = 0$ comes back twice: in LP's first stage (ACF) and for gross output (GNR).

Selection (Olley and Pakes 1996)

Firms exit when productivity falls below a threshold that depends on their state:

$$\chi_{jt} = \mathbf{1}\{\omega_{jt} \geq \underline{\omega}_t(k_{jt}, a_{jt})\}.$$

- Larger capital stocks mean higher continuation values, so $\underline{\omega}_t$ is decreasing in k: big plants survive worse draws.
- Among survivors, $\mathbb{E}[\omega_{jt} \mid k_{jt}, \chi_{jt} = 1]$ is decreasing in k, which biases $\hat{\beta}_k$ down.
- OP's telecom data: a balanced panel throws away about 65% of observations and gives $\hat{\beta}_k = 0.173$; the full sample with their correction gives 0.355.
- Same logic whenever selection is on productivity: exporters (Melitz 2003), entrants, survivors of a trade shock.

Measuring Output: Revenue Is Not Quantity

Most producer data record sales and expenditures. With an industry deflator $\bar{p}_t$:

$$r_{jt} - \ln \bar{p}_t = f(\mathbf{x}_{jt}) + \omega_{jt} + \varepsilon_{jt} + \underbrace{(\ln p_{jt} - \ln \bar{p}_t)}_{\text{omitted price}}.$$

- If firms set prices, $\ln p_{jt}$ falls as output rises, so it is correlated with every input. Klette and Griliches (1996): the estimated elasticities are scaled down by $1/\mu$ (next slide).
- Input expenditures have the same problem when input prices vary across firms.
- Even with quantities: multi-product producers, quality differences, units that are not comparable across firms.
- We come back to this when we estimate markups, where it matters most.

TFPR Is Not the Revenue Residual

Klette and Griliches (1996): $y_{jt} = f(\mathbf{x}_{jt}) + \omega_{jt}$ (drop ε) and CES demand $y_{jt} - \bar{y}_t = -\eta\,(\ln p_{jt} - \ln \bar{p}_t) + \delta_{jt}$, with demand shifter δ_{jt} and $\eta > 1$, so $\mu = \eta/(\eta - 1)$. Substitute the price out of deflated revenue $\tilde{r}_{jt} \equiv r_{jt} - \ln \bar{p}_t = y_{jt} + \ln p_{jt} - \ln \bar{p}_t$:

$$\tilde{r}_{jt} = \underbrace{\tfrac{\eta-1}{\eta}\, f(\mathbf{x}_{jt})}_{\text{elasticities } \theta^X/\mu} + \tfrac{1}{\eta}\,\bar{y}_t + \underbrace{\tfrac{\eta-1}{\eta}\,\omega_{jt} + \tfrac{1}{\eta}\,\delta_{jt}}_{\omega^R_{jt}\text{: revenue residual}},$$

$$\text{TFPR}_{jt} \equiv \tilde{r}_{jt} - f(\mathbf{x}_{jt}) = \omega_{jt} + \ln p_{jt} - \ln \bar{p}_t = \omega^R_{jt} - \tfrac{1}{\eta}\big(f(\mathbf{x}_{jt}) - \bar{y}_t\big).$$

- The revenue regression recovers θ^X/μ, not θ^X: the Klette–Griliches bias.
- The gap grows with the firm's inputs at rate $1/\eta$, a demand parameter: big firms look better by ω^R.
- Starkest case (Hsieh–Klenow): CRS, flexible inputs, common input prices $\mathbf{w}_t$. Then $mc = c(\mathbf{w}_t)/\Omega$ and $p = \mu\, mc$, so $\text{TFPR}_{jt} = \ln \mu + \ln c(\mathbf{w}_t) - \ln \bar{p}_t$: the same for every firm. With $\eta = 4$, 10% more TFPQ raises ω^R by 7.5% and a 10% demand shock raises it by 2.5%; neither moves TFPR.

De Loecker–Syverson (2021), fns. 6 and 42. De Loecker (2011) estimates η inside the production function (later).

Foster, Haltiwanger and Syverson (2008)

Eleven physically homogeneous products (ready-mixed concrete, block ice, sugar, plywood, ...), US Census of Manufactures 1977–97, 17,669 plant-years. Prices and quantities are both observed.

Two benchmarks bracket the data. Price taking: corr$(p, \text{TFPQ}) = 0$ and TFPR $=$ TFPQ. Hsieh–Klenow (previous slide): corr$(p, \text{TFPQ}) = -1$ and TFPR does not vary.

Correlations	Price	TFPR	TFPQ
Price	1.00		
TFPR	0.16	1.00	
TFPQ	-0.54	0.75	1.00
s.d.	0.18	0.22	0.26
1-yr persistence	0.82	0.77	0.79

FHS Tables 1 and 3; product-year effects removed. Demand shocks: persistence 0.91.

- The data sit in between: more physically productive plants charge lower prices, so TFPR understates TFPQ dispersion.
- Entrants have higher TFPQ than incumbents but charge lower prices: revenue measures understate the entrant advantage.
- Exit depends on TFPQ and on demand. Selection is on profitability, not just productivity.
- Demand shocks are more persistent than productivity.

Classic Fixes and What They Assume

Fix	Assumes	Problem in practice
Fixed effects	$\omega_{jt} = \omega_j$	tiny $\widehat{\beta}_k$ and low returns to scale; differencing amplifies measurement error (OP within: $\widehat{\beta}_k = 0.067$)
Input-price IV	firm-level w^X observed and exogenous	rarely observed; quality and location confounds
Cost shares / index numbers	static FOCs for *all* inputs, CRS, no ex post shock	capital and labor have adjustment costs; with revenue shares also needs $\mu = 1$

- The modern estimators relax some of these and add others: a timing assumption, a Markov process for ω, and either linearity or a scalar unobservable.
- ACF's advice: be explicit about which assumptions you are making, and whether they fit the industry.

One Equation, Three Families

One Equation Behind Every Estimator

Substitute the Markov process into the production function:

$$y_{jt} = f(\mathbf{x}_{jt}) + g(\omega_{jt-1}) + \xi_{jt} + \varepsilon_{jt}, \qquad \mathbb{E}[\xi_{jt} + \varepsilon_{jt} \mid \mathbf{z}_{jt}] = 0,$$

with instruments $\mathbf{z}_{jt} \subseteq \mathcal{I}_{jt-1}$, e.g. $(k_{jt}, l_{jt-1}, m_{jt-1})$. The estimators differ in how they replace ω_{jt-1}:

Family	ω_{jt-1} replaced by	Key extra assumption
Dynamic panel	$y_{jt-1} - f(\mathbf{x}_{jt-1}) - \varepsilon_{jt-1}$	g linear
Proxy / control function	$h_{t-1}(\mathbf{o}_{jt-1})$, an inverted input demand	scalar unobservable, monotone input demand
FOC / share form	$y_{jt-1} - f(\mathbf{x}_{jt-1}) + \ln(\alpha^X_{jt-1}/\theta^X_{jt-1})$	static FOC for X; price taking (or its cost-share version)

Framing from Jaumandreu (2025); the FOC row starts with Doraszelski–Jaumandreu (2013). With a linear g and correct FOCs, all three should agree up to sampling error. Disagreement is informative.

Dynamic Panel (Blundell and Bond 2000)

With $\omega_{jt} = \rho\,\omega_{jt-1} + \xi_{jt}$, quasi-difference:

$$y_{jt} - \rho\, y_{jt-1} = f(\mathbf{x}_{jt}) - \rho f(\mathbf{x}_{jt-1}) + \xi_{jt} + \varepsilon_{jt} - \rho\,\varepsilon_{jt-1},$$

instrumented with $(k_{jt}, k_{jt-1}, l_{jt-1}, m_{jt-1})$ and lagged prices.

- ✓ No model of input demand: allows unobserved input-price shocks to every input, demand shocks, any conduct, and fixed effects.
- ✗ Needs g linear; selection is hard to handle; ε is not separated from ω.
- ✗ In practice often implausible: in Compustat a Cobb–Douglas dynamic panel gives $\hat{\beta}_K < 0$ and variable returns above 1.1 (Jaumandreu 2025).
- For a flexible input, lagged inputs are relevant only if something persistent moves input demand besides ω: firm-specific input prices, or demand differences under market power (Bond et al. 2021).

Olley and Pakes (1996)

Olley–Pakes: Deregulating Telecom Equipment

- AT&T's network monopoly extended to equipment: it bought almost everything from its subsidiary Western Electric, and connecting outside equipment was restricted.
- Two shocks: FCC certification of outside equipment (late 1970s), and the AT&T divestiture (decreed 1982, effective 1984). Western Electric's share of Bell company purchases fell from 92% (1982) to 58% (1986).
- Massive turnover: plants rose from 133 (1963) to 584 (1987); 79% of plants active in 1987 entered after 1972, and 60% of 1972 plants had exited.
- Questions: did productivity rise, and through which channel, within plants or reallocation across them? Answering needs production function estimates that handle simultaneity and selection.

Source: Olley and Pakes (1996), Tables I–IV.

The OP Model

Incumbents decide each period whether to exit (receiving a scrap value $\underline{V}$), and if not, choose labor and investment:

$$V_t(\omega_t, a_t, k_t) = \max\left\{\underline{V},\ \sup_{i_t \geq 0}\ \pi_t(\omega_t, a_t, k_t) - c(i_t) + \beta\,\mathbb{E}\left[V_{t+1}(\omega_{t+1}, a_{t+1}, k_{t+1}) \mid \mathcal{J}_t\right]\right\}$$

with $K_{t+1} = (1-\delta_K)K_t + i_t$, $a_{t+1} = a_t + 1$, and β the discount factor.

- Market structure and factor prices are common to all firms, so they are absorbed into the t subscripts (assumed Markov perfect equilibrium).
- Policies: exit if $\omega_t < \underline{\omega}_t(a_t, k_t)$; invest $i_t = i_t(\omega_t, a_t, k_t)$.
- Key result (Pakes 1994): if $i_t > 0$, investment is strictly increasing in ω_t, so it can be inverted:

$$\omega_{jt} = h_t(i_{jt}, a_{jt}, k_{jt}).$$

- This rests on ω being the only firm-specific unobserved state: no unobserved differences in demand, input prices, or adjustment costs.

OP Estimation in Three Steps

1. First stage (partially linear, Robinson 1988):

$$y_{jt} = \beta_l \, l_{jt} + \Phi_t(i_{jt}, a_{jt}, k_{jt}) + \varepsilon_{jt}, \qquad \Phi_t = \beta_0 + \beta_a a + \beta_k k + h_t(i, a, k).$$

   Recovers $\hat{\beta}_l$ and $\hat{\Phi}_{jt}$ (polynomial or kernel). Capital and age are inside Φ.

2. Survival: $\mathcal{P}_{jt} = \mathbb{P}(\chi_{jt+1} = 1 \mid i_{jt}, a_{jt}, k_{jt})$, which indexes the exit threshold.

3. Second stage (NLLS):

$$y_{jt+1} - \hat{\beta}_l \, l_{jt+1} = \beta_a a_{jt+1} + \beta_k k_{jt+1} + \tilde{g}(\mathcal{P}_{jt}, \; \hat{\Phi}_{jt} - \beta_a a_{jt} - \beta_k k_{jt}) + \xi_{jt+1} + \varepsilon_{jt+1}.$$

- k_{jt+1} was chosen at t, so it is uncorrelated with ξ_{jt+1}. Labor responds to ξ_{jt+1}, which is why β_l comes from step 1.
- Observations with zero investment are dropped (8% in OP's data; far more elsewhere).

OP Results

	Balanced panel		Full sample		OP
	OLS	Within	OLS	Within	Kernel
Labor	.851	.728	.693	.629	.608
	(.039)	(.049)	(.019)	(.026)	(.027)
Capital	.173	.067	.304	.150	.355
	(.034)	(.049)	(.018)	(.026)	(.058)
Obs.	896	896	2592	2592	1758

- Labor falls and capital rises as simultaneity and selection are corrected, in the directions theory predicts.
- Within estimators kill the capital coefficient: productivity is not a fixed effect during a restructuring.
- Overidentification checks (adding k or a inside $\tilde{g}$) do not reject.

Source: Olley and Pakes (1996), Table VI, columns (1)–(4) and (9).

The Olley–Pakes Decomposition

With $\Omega_{jt} = e^{\omega_{jt}}$ and output shares λ_{jt}, industry productivity is

$$\Omega_t \equiv \sum_j \lambda_{jt}\,\Omega_{jt} = \bar{\Omega}_t + \sum_j (\lambda_{jt} - \bar{\lambda}_t)(\Omega_{jt} - \bar{\Omega}_t),$$

the unweighted mean plus a covariance between size and productivity.

Year	Ω_t	$\bar{\Omega}_t$	Covariance
1975	0.72	0.66	0.06
1978	0.92	0.80	0.12
1981	1.11	0.76	0.35
1984	0.90	0.83	0.07
1987	0.97	0.66	0.32

OP (1996), Table XI.

- The unweighted mean barely moves; the gains come from output moving toward more productive plants.
- Across studies the covariance term explains 20–40% of changes in share-weighted productivity (De Loecker–Syverson 2021).
- This is revenue productivity.
- Caveats: share-weighted productivity is not the theoretically correct aggregate (Petrin–Levinsohn 2012); entry and exit need the dynamic version (Melitz–Polanec 2015).

Levinsohn–Petrin and Ackerberg–Caves–Frazer

Levinsohn and Petrin (2003): Invert Materials Instead

- Use an intermediate input as the proxy: $m_{jt} = \mathbb{M}_t(k_{jt}, \omega_{jt})$, so $\omega_{jt} = h_t(k_{jt}, m_{jt})$.
- Why:
  - investment is often zero (those observations are dropped) and lumpy (flat spots in $i(\omega)$);
  - materials are static, so monotonicity is a property of f rather than of a dynamic equilibrium;
  - unobserved capital adjustment costs no longer break the inversion.
- Monotonicity from the implicit function theorem on the FOCs for (m, l):

$$\text{sign}\left(\frac{\partial m}{\partial \omega}\right) = \text{sign}\left(f_{ml} f_{l\omega} - f_{ll} f_{m\omega}\right),$$

  when the (m, l) Hessian is negative definite.

Share of non-zero observations, Chile

Industry	Investment	Materials
Food (311)	42.7	99.8
Metals (381)	44.8	99.9
Textiles (321)	41.2	99.9
Wood (331)	35.9	99.7

LP (2003), Table 2. OP's industry: 8% zeros.

The ACF Critique: Functional Dependence

LP's first stage, $y_{jt} = \beta_l l_{jt} + \Phi_t(k_{jt}, m_{jt}) + \varepsilon_{jt}$, needs variation in labor that is not explained by (k, m) (Robinson 1988):

$$\mathbb{E}\Big[\big(l_{jt} - \mathbb{E}[l_{jt} \mid k_{jt}, m_{jt}]\big)^2\Big] > 0.$$

- But if labor is chosen like materials, $l_{jt} = \ell_t(k_{jt}, \omega_{jt}) = \ell_t(k_{jt}, h_t(k_{jt}, m_{jt}))$: no variation left.
- The simultaneity algebra again, now in LP's first stage. Cobb–Douglas with a price-taking FOC for m, inverted for ω and plugged back in:

$$y_{jt} = \ln\left(\frac{1}{\beta_m}\right) + \ln\left(\frac{w_t^M}{p_t}\right) + m_{jt} + \varepsilon_{jt}.$$

  β_l does not appear at all.
- With polynomials you will still get a number. It just is not a consistent estimate of β_l.
- The same argument applies to OP's investment proxy.

What Would Break the Functional Dependence?

ACF search for data-generating processes consistent with LP that still identify β_l in the first stage. Only knife-edge cases work:

1. iid optimization error in labor, but not in materials (workers calling in sick, with materials ordered beforehand);
2. an iid shock to wages or demand realized after m is chosen and before l;
3. (OP only) labor chosen at $t-b$, on ω_{jt-b}, and non-dynamic, while investment is chosen at t.

What does not help:

- observed firm-level input prices (they enter Φ_t too);
- dynamic labor (l_{jt-1} becomes a state in both input demands);
- classical measurement error.

The ACF Procedure

Invert the demand for materials conditional on labor, $m_{jt} = \tilde{\mathbb{M}}_t(k_{jt}, l_{jt}, \omega_{jt})$, so $\omega_{jt} = h_t(k_{jt}, l_{jt}, m_{jt})$.

1. First stage identifies nothing but nets out ε: $\quad y_{jt} = \Phi_t(k_{jt}, l_{jt}, m_{jt}) + \varepsilon_{jt}$.
2. For trial β: $\omega_{jt}(\beta) = \hat{\Phi}_{jt} - \beta_k k_{jt} - \beta_l l_{jt}$; regress on $\omega_{jt-1}(\beta)$ to get $\xi_{jt}(\beta)$; solve

$$\mathbb{E}\left[\xi_{jt}(\beta) \otimes (1,\ k_{jt},\ l_{jt-1},\ \hat{\Phi}_{jt-1})'\right] = 0.$$

- Allows dynamic labor, labor chosen before materials, and serially correlated unobserved wage shocks.
- Does not allow unobserved materials prices or demand shocks.
- Meant for value added, or gross output that is Leontief in m. Next: why not general gross output.
- Wooldridge (2009): both stages as one GMM system; efficient, easy robust SEs.

Serially correlated unobserved cost shocks allowed?

	to k	to l	to m
OP	✗	✗	✗
LP	✓	✗	✗
ACF	✓	✓	✗
Dynamic panel	✓	✓	✓

ACF Monte Carlo

Leontief in materials; true $(\beta_l, \beta_k) = (0.6, 0.4)$; 1000 firms $\times$ 10 periods; no measurement error in m.

	ACF		LP	
DGP	$\hat{\beta}_l$	$\hat{\beta}_k$	$\hat{\beta}_l$	$\hat{\beta}_k$
1: serially correlated wages, l chosen at $t-b$	0.600	0.399	0.000	1.121
2: optimization error in l (LP's case)	0.600	0.400	0.600	0.399
3: both	0.596	0.406	0.473	0.588

- DGP 1: functional dependence in action, as LP's labor coefficient collapses to zero.
- DGP 2: LP is consistent and more precise; ACF pays an efficiency cost for robustness.
- OLS gives $\hat{\beta}_l \approx 0.85$–$0.95$. OP with firm-specific capital adjustment costs gives $\hat{\beta}_l \approx 0.82$.
- Adding measurement error to m moves LP much more than ACF.

Source: Ackerberg, Caves and Frazer (2015), Tables I–II.

Gross Output

Gross Output Is Not Identified by These Moments (GNR 2020)

Put materials back in: $y_{jt} = f(k_{jt}, l_{jt}, m_{jt}) + \omega_{jt} + \varepsilon_{jt}$, with m flexible. After conditioning on the instruments, what still moves materials?

$$m_{jt} = \mathbb{M}(k_{jt},\ l_{jt},\ g(h(\mathbf{o}_{jt-1})) + \xi_{jt} - d_t), \qquad d_t = \ln(w_t^M / p_t).$$

Only ξ_{jt}, which is orthogonal to every instrument by construction, and the aggregate price ratio d_t.

- Theorem 1: with $d_t = d$, for every $a \in (0, 1)$

$$\tilde{f} = (1-a)\, f + a\, \Phi, \qquad \tilde{g}(x) = ad + (1-a)\, g\Big(\frac{x - ad}{1-a}\Big)$$

  satisfies exactly the same moment conditions. The same holds for investment proxies and for dynamic panel (their App. O1).
- Time-series price variation identifies in theory but not in practice. In their Monte Carlo the estimated θ^M is about 0.9–1.0 against a truth of 0.65, unless price variation is 10 times the baseline and $T \geq 20$.
- What you need is an excluded shifter of materials demand that varies across firms ($\sigma_d^2 > 0$: firm-level input prices), or known returns to scale instead (FGT 2019).

GNR's Fix: Use the First-Order Condition

Under price-taking in output and materials, take logs of the FOC $p_t \, \partial F/\partial M \, e^{\omega} \, \mathcal{E} = w_t^M$ and subtract the production function:

$$\ln \alpha_{jt}^M = \ln \mathcal{E} + \ln \theta^M(k_{jt}, l_{jt}, m_{jt}) - \varepsilon_{jt}, \qquad \mathcal{E} = \mathbb{E}[e^{\varepsilon}].$$

1. A nonparametric share regression identifies $\theta^M(\cdot)$ and ε_{jt}.
2. Integrate the elasticity: $\int \theta^M \, dm = f(k, l, m) + \mathcal{C}(k, l)$.
3. The remainder obeys a dynamic panel structure:

$$y_{jt} \equiv y_{jt} - \varepsilon_{jt} - \int \theta^M dm_{jt} = -\mathcal{C}(k_{jt}, l_{jt}) + \omega_{jt},$$

with moments $\mathbb{E}[\xi_{jt} \mid k_{jt}, l_{jt}, y_{jt-1}, k_{jt-1}, l_{jt-1}] = 0$.

A nonparametric version of Griliches–Ringstad (1971): index-number logic for the flexible input, lags for the rest. Estimation uses polynomial sieves with closed-form integrals and a bootstrap.

GNR Estimates: Chile and Colombia

	Colombia, all mfg.		Chile, all mfg.		Chile, food (311)	
Elasticity	GNR	OLS	GNR	OLS	GNR	OLS
Labor	.35	.26	.38	.20	.28	.17
Capital	.14	.06	.16	.09	.11	.05
Intermediates	.54	.72	.55	.77	.67	.83
Sum	1.04	1.04	1.09	1.06	1.05	1.05

- OLS overstates the intermediate-input elasticity (by 34% on average) and understates labor and capital.
- OLS also understates productivity dispersion (Colombia: 95/5 ratio of 2.36 under GNR vs 1.94 under OLS) and persistence (0.64 vs 0.53).
- Robust to fixed effects, to treating part of intermediates as non-flexible, and to large adjustment costs in materials (Monte Carlo).

Source: Gandhi, Navarro and Rivers (2020), Tables 2–3.

Leontief in Materials: Why It Is Useful

Structural value added: $Y_{jt} = \min\left\{F(K_{jt}, L_{jt})\, e^{\omega_{jt}},\ \gamma M_{jt}\right\} e^{\varepsilon_{jt}}$; a producing firm sets $\gamma M = F(K, L)\, e^{\omega}$.

- Identification: materials drop out of what you estimate, $y = f(k, l) + \omega + \varepsilon$. Yet they remain a perfect proxy, $m = f(k, l) + \omega - \ln\gamma$, monotone in ω given (k, l). GNR's non-identification result has no flexible input to bite on. This is the case ACF have in mind.
- Plausible at a disaggregated level, where materials are a recipe: steel mills (Hendel–Spiegel 2014), brewing (De Loecker–Scott 2016), oil wells (Asker–Collard-Wexler–De Loecker 2019), and cigarettes, with 686 mg of leaf per 1000 mg cigarette and s.d. 30 mg across Chinese manufacturers (Rubens 2023).
- Markups: at the kink materials have no FOC, so use labor. But marginal cost includes materials:

$$mc = \frac{w^M}{\gamma} + \frac{w^L L}{\theta^L Y} \quad \Longrightarrow \quad \mu = \left(\alpha^M + \frac{\alpha^L}{\theta^L}\right)^{-1}.$$

  The naive θ^L/α^L overstates μ: with $\alpha^M = 0.6$, $\alpha^L = 0.2$, $\theta^L = 0.6$, it gives 3.0 against the true 1.07 (De Loecker–Scott; Rubens 2023, eq. 4d).
- Cautions: idle capacity breaks the Leontief equality (GNR's critique, ACF App.); estimated labor–materials substitution elasticities are well above zero (Doraszelski–Jaumandreu 2018; Edmond–Midrigan–Xu calibrate 0.5).

Endogenous Productivity

Endogenous Productivity: Firm Actions Belong in g

Exporting, R&D, or trade protection at $t-1$ shift the conditional mean of productivity: $\omega_{jt} = g(\omega_{jt-1}, a_{jt-1}) + \xi_{jt}$, with $a_{jt-1} \in \mathcal{J}_{jt-1}$. The moments are unchanged; g just gains an argument.

Suppose the truth is $\omega_{jt} = \omega_{jt-1} + \gamma\, e_{jt-1} + \xi_{jt}$ with export status e, but g is treated as exogenous (De Loecker 2013). With $\tilde{y} = y - \hat{\beta}_l l$:

$$\Delta\tilde{y}_{jt} = \beta_k \Delta k_{jt} + \underbrace{\gamma\, e_{jt-1} + \xi_{jt}}_{\text{error}} + \Delta\varepsilon_{jt} \quad \Longrightarrow \quad \text{plim}\, \hat{\beta}_k = \beta_k + \gamma\, \frac{\mathbb{C}(\Delta k_{jt}, e_{jt-1})}{\mathbb{V}(\Delta k_{jt})} > \beta_k.$$

- Exporters invest more, so $\hat{\beta}_k$ is too high, and a DID on $\hat{\omega}$ subtracts too much of export starters' capital growth. Slovenia: $\hat{\beta}_k$ is about 30% higher under an exogenous g, and the DID misses 1.5 pp of a 4.1% learning-by-exporting effect in the entry year.
- Estimating $\hat{\omega}$ with an exogenous g and then regressing it on e tests what step 1 assumed away.
- R&D (Doraszelski–Jaumandreu 2013, Spain): the exogenous process is rejected in all 9 industries; R&D makes productivity riskier (larger $\hat{\xi}^2$ in every industry, while physical investment does not); the net return is about 40%, roughly twice that on physical capital.

Markups from Production Data

Three Ways to Measure a Markup

$\mu_{jt} \equiv p_{jt}/mc_{jt}$. Marginal cost is rarely observed, so something has to stand in for it.

- Accounting: revenue over total cost. Right only if average cost equals marginal cost (CRS, no fixed costs).
- Demand approach (NEIO; Weeks 2–7): estimate demand elasticities, assume a conduct model, and back out mc from the pricing FOC.
  - needs product-level prices and quantities, instruments, and a model of competition;
  - delivers counterfactuals (mergers), which the production approach cannot.
- Production approach (Hall 1988; De Loecker–Warzynski 2012): cost minimization in one frictionless variable input.
  - no demand system and no conduct assumption;
  - needs an output elasticity, hence a production function;
  - works across many industries at once, so it has driven the macro debate on market power.
- They rest on different assumptions, so agreement is informative: De Loecker and Scott (2016) find the two approaches agree for US brewing.

De Loecker and Warzynski (2012): The Markup Formula

Minimize cost given dynamic inputs (capital) and productivity:

$$\min_{\{X\}_{X\in\mathcal{V}}} \sum_{X\in\mathcal{V}} w^X X \quad \text{s.t.} \quad F(\mathbf{x}, K)\, e^{\omega} \geq Y \quad \Longrightarrow \quad w^X = mc \cdot \frac{\partial Y}{\partial X}.$$

Multiply by X/Y, then multiply and divide by p:

$$\theta^X_{jt} = \frac{p_{jt}}{mc_{jt}} \cdot \frac{w^X_{jt} X_{jt}}{p_{jt} Y_{jt}} = \mu_{jt}\, \alpha^X_{jt} \qquad \Longrightarrow \qquad \boxed{\mu_{jt} = \frac{\theta^X_{jt}}{\alpha^X_{jt}}}$$

- Requires: cost minimization; X free of adjustment costs; firm is a price taker in the market for X; prices set period by period.
- Does not require: a demand system, a conduct model, constant returns, or the user cost of capital (we condition on K).
- The formula is old (Hall 1988 assumed μ constant across firms). What is new is firm-level θ^X_{jt} from a carefully estimated production function.

Implementing DLW

- Estimate f by ACF, with extra input-demand shifters $\mathbf{d}_{jt}$ in the proxy: $m_{jt} = \mathbb{M}_t(k_{jt}, \omega_{jt}, \mathbf{d}_{jt})$.
  - DLW include input prices and export status. De Loecker et al. (2016) add output prices, market shares, product dummies, and tariffs.
  - This is the only restriction on competition: everything that shifts input demand across firms must be in **d**.
  - Monotonicity under imperfect competition: more productive firms must not set "inordinately" higher markups (Melitz–Levinsohn).
- Translog: $\widehat{\theta}^L_{jt} = \widehat{\beta}_l + 2\widehat{\beta}_{ll}\, l_{jt} + \widehat{\beta}_{lk}\, k_{jt}$. Under Cobb–Douglas θ is constant, so all variation in the share becomes variation in the markup.
- Use planned rather than realized output in the share: $\widehat{\alpha}^X_{jt} = w^X_{jt} X_{jt} / (p_{jt} Y_{jt}\, e^{-\widehat{\varepsilon}_{jt}})$.
- Then project: $\ln \widehat{\mu}_{jt} = \delta_0 + \delta_1 \text{ export}_{jt} + \text{controls} + e_{jt}$.

Multi-Product Firms (De Loecker, Goldberg, Khandelwal, Pavcnik 2016)

Indian manufacturers, 1989–2003: product quantities and unit-value prices, but inputs only at the firm level, as deflated expenditures $\tilde{\mathbf{x}}_{jt}$. Products n, input shares ϕ_{jnt}.

1. Estimate f on single-product firms (selection-corrected), proxying input prices by $\ln \tilde{w} = \mathcal{W}(\ln p, \text{market share}, \text{product})$.
2. ω_{jt} is common to the firm's products, so physical output pins down each product's input share:

$$\hat{y}_{jnt} = f(\tilde{\mathbf{x}}_{jt} + \ln \phi_{jnt} - \ln \hat{\tilde{w}}_{jnt}) + \omega_{jt}, \qquad \sum_n \phi_{jnt} = 1.$$

   Cobb–Douglas with returns Γ: $\phi_{jnt} \propto \hat{\tilde{w}}_{jnt} \hat{Y}_{jnt}^{1/\Gamma}$. Splitting inputs by revenue is right only if markups are equal across products.
3. $\mu_{jnt} = \theta_{jnt}^M p_{jnt} Y_{jnt} / (\phi_{jnt} w_{jt}^M M_{jt})$, and $mc_{jnt} = p_{jnt}/\mu_{jnt}$.

India 1989–97, %	output tariffs	input tariffs	total
$\Delta \ln p$	-9.7	-8.4	-18.1
$\Delta \ln mc$	-2.9	-27.8	-30.7
$\Delta \ln \mu$	-6.8	$+19.4$	$+12.6$

Tariffs fell 62 pp (output) and 24 pp (input). Cheaper inputs were kept as markup: pass-through about 0.3. Median markup 1.34 (mean 2.70). The markup total is not significant (s.e. 11.9). Closed form and split derived here (eqs. 23–24, App. B, Table IX).

One Markup, Many Inputs

If the FOC holds for every variable input, $\theta^X = \mu\,\alpha^X$ for all $X \in \mathcal{V}$. Sum over X, with $\nu_{jt} \equiv \sum_{X\in\mathcal{V}} \theta^X_{jt}$ the short-run scale elasticity (ν here is not a random-coefficient draw):

$$\nu_{jt} = \mu_{jt}\,\alpha^{\mathcal{V}}_{jt} \qquad \Longrightarrow \qquad \boxed{\mu_{jt} = \nu_{jt}\,\frac{p_{jt}Y_{jt}}{VC_{jt}}}, \qquad \frac{\theta^X_{jt}}{\nu_{jt}} = \frac{w^X_{jt}X_{jt}}{VC_{jt}}.$$

- The markup is short-run returns to scale times revenue over variable cost. Equivalently, $mc = AVC/\nu$ (Jaumandreu 2025). With CRS and all inputs variable, this is the accounting markup.
- What you really need is ν, not each θ^X.
- With two flexible inputs the markup is overidentified: labor and materials give the same μ iff

$$\frac{\theta^L_{jt}}{\theta^M_{jt}} = \frac{w^L_{jt}L_{jt}}{w^M_{jt}M_{jt}}.$$

  This is exactly Raval's (2023) test, and it is what fails in the data.

Combining DLW's formula across inputs; derived here.

DLW Results: Slovenian Manufacturing, 1994–2000

Median markups	
Hall (1986)	1.03
Klette (1999)	1.12
Cobb–Douglas, value added	1.17
Translog, value added	1.28
Gross output, labor FOC	1.26
Gross output, materials FOC	1.22
Single markup, levels	1.16
Single markup, first differences	1.11

DLW Table 2; s.d. of firm markups about 0.5.

- 7,915 firms. Controlling for productivity raises the estimates relative to Hall and Klette.
- Using capital's "FOC" gives 1.5–1.6. That wedge includes adjustment costs, which is why capital is not used.
- Exporters charge higher markups: about 8% (0.13–0.18 in levels). This falls to 2% once productivity is controlled, so most of the gap is lower marginal cost.
- Markups rise when firms start exporting.

One FOC, Two Unknowns

The flexible-input FOC, $\ln \alpha_{jt}^{M} = \ln \theta_{jt}^{M} - \ln \mu_{jt} + \ln \mathcal{E} - \varepsilon_{jt}$, is one equation in two unknowns.

- GNR read it left to right with $\mu \equiv 1$ (price taking): the share regression identifies θ^{M}. Under market power it recovers θ^{M}/μ instead.
- DLW read it right to left: take θ^{M} from the production function and back out μ. The markup is identified by whatever identifies θ^{M}: an excluded shifter of input demand, here serially correlated firm-level input prices (DLW p. 2446). Unobserved, those prices would also break the proxy (Flynn–Gandhi–Traina 2019).
- Flynn–Gandhi–Traina (2019) use a shape restriction instead. With Cobb–Douglas and $\pi_x \equiv \partial \mathbb{E}[x_{jt} \mid k_{jt}, \mathbf{o}_{jt-1}]/\partial k_{jt}$, timing gives one equation, $\pi_y = \theta^{K} + \theta^{M}\pi_m$. Constant returns close it:

$$\theta^{M} = \frac{1-\pi_y}{1-\pi_m}, \qquad \mu = (1-\theta^{K})\,\frac{pY}{VC}.$$

  This fails when $\pi_m = 1$, which (with a constant markup) happens exactly when $\mu = 1$: GNR's case. It also needs quantity output. With revenue and a constant markup, planned revenue is $\ln(\mu/\theta^{M}) + \ln w^{M} + m$, so $\pi_r = \pi_m$ and the same moments return $\widehat{\theta}^{M} = 1$ and $\widehat{\mu} = pY/VC = \mu/\theta^{M}$: Bond et al.'s problem in CRS form.

Derived here from GNR (2020), DLW (2012) and FGT (2019).

Where the Production Approach Breaks

Estimate the Demand Side Too (De Loecker 2011)

Belgian textiles, 1994–2002: no prices, but segment output $\bar{y}_{st}$ (including imports), EU quota protection $qr_{jt} \in [0,1]$, and product dummies $\mathbf{D}_j$. Take the TFPR slide's equation with $\eta \to \eta_s$ and $\delta_{jt} = \mathbf{D}_j'\boldsymbol{\kappa} + \tau\, qr_{jt} + \tilde{\delta}_{jt}$:

$$\tilde{r}_{jt} = \sum_X \underbrace{\frac{\theta^X}{\mu_s}}_{\beta_X} x_{jt} + \underbrace{\frac{1}{\eta_s}}_{\beta_s} \bar{y}_{st} + \frac{1}{\eta_s}(\tau\, qr_{jt} + \mathbf{D}_j'\boldsymbol{\kappa}) + \frac{\eta_s - 1}{\eta_s}\,\omega_{jt} + \tilde{\varepsilon}_{jt},$$

$$\omega_{jt} = g(\omega_{jt-1}, qr_{jt-1}) + \xi_{jt}, \qquad \omega_{jt} = h_t(k_{jt}, m_{jt}, qr_{jt}, \bar{y}_{st}, \mathbf{D}_j).$$

- The coefficient on market size is the Lerner index: $\hat{\mu}_s = 1/(1-\hat{\beta}_s)$, $\hat{\theta}^X = \hat{\mu}_s\,\hat{\beta}_X$. Quotas shift demand at once (τ), productivity only with a lag (g).
- With demand controls: $\hat{\eta}$ 2.86 $\to$ 3.85, $\hat{\mu}$ 1.54 $\to$ 1.35, returns to scale 1.3 $\to$ 1.11 (revenue OLS: 0.95).
- Removing all quotas raises productivity 2.1%, not 8.6% (over three years: 4.6%, not 13.5%).
- CES fixes μ_s, so liberalization cannot move markups; β_X is the next slide's revenue elasticity.

Revenue Elasticities Carry No Markup Information (Bond et al. 2021)

With revenue data we can at best learn the revenue elasticity. With $R = pY$:

$$\theta_R^X \equiv \frac{\partial r_{jt}}{\partial x_{jt}} = \frac{\partial \ln R}{\partial \ln Y}\,\theta_{jt}^X = \frac{MR_{jt}}{p_{jt}}\,\theta_{jt}^X \quad \Longrightarrow \quad \frac{\theta_R^X}{\alpha_{jt}^X} = \frac{MR_{jt}}{p_{jt}}\,\mu_{jt} = \frac{MR_{jt}}{mc_{jt}} = 1.$$

- This holds under profit maximization for any demand system and market structure (their App. A.1: Bertrand and Cournot).
- Klette–Griliches said revenue biases elasticities down. This is the strong form: the firm's own revenue elasticity contains no markup information.
- Revenue data cannot even identify θ^X nonparametrically. m and p solve the same problem, $m = m^*(k, l, \omega, \delta, w^M)$ and $p = p^*(k, l, \omega, \delta, w^M)$, which is ACF-style functional dependence.
- With quantity data, identification needs persistent firm differences in w^M *or* in demand. Market power turns demand heterogeneity into instrument relevance.
- Their bottom line: levels need firm-level output prices. Mean differences between groups with a common technology (exporters) survive; trends need a stable θ.

...But Dispersion and Trends Survive (De Ridder, Grassi, Morzenti 2026)

Estimators never deliver firm j's own revenue elasticity; they deliver a weighted average across firms. Cobb–Douglas, log variable input v_{jt}, lagged input as instrument:

$$\hat{\theta} = \theta + \frac{\mathbb{E}[\ln p_{jt}\, v_{jt-1}]}{\mathbb{E}[v_{jt}\, v_{jt-1}]}, \qquad \hat{\mu}^R_{jt} = \underbrace{\mathbb{E}\left[\frac{\varpi_{jt}}{\mu_{jt}}\right]}_{\text{common constant}} \mu_{jt}, \qquad \varpi_{jt} = \frac{v_{jt}\, v_{jt-1}}{\mathbb{E}[v_{jt}\, v_{jt-1}]}.$$

- So corr($\ln \hat{\mu}^R, \ln \mu$) $= 1$ while the level is uninformative (its mean is about 1). With identical demand elasticities (CES, atomistic firms) every firm has $\hat{\mu}^R = 1$: the Bond et al. case.
- Translog: $\hat{\mu}^R_{jt} = (1 + f(v_{jt}))\, \mu_{jt}$. The correlation stays high when prices are close to orthogonal to the instruments: 0.94 in an Atkeson–Burstein simulation.
- France 2009–19, with firm prices from unit values: corr(revenue markup, quantity markup) = 0.63 within sectors (0.80 in first differences). Mean markup is 1.08 from revenue vs 1.45 from quantities, but 2010–19 trends match, and the aggregation method moves the level, not the trend.
- Rule: levels need prices. Dispersion and trends survive revenue data if technology is common across firms (and stable over time).

When Productivity Cannot Be Inverted (Doraszelski and Li 2025)

The proxy assumes invertibility: $\mathbb{E}[\omega_{jt} \mid \mathbf{o}_{jt}] = \omega_{jt}$. It fails with demand shocks δ_{jt}, unobserved rivals, changes in conduct (ownership), or unobserved input prices.

- Test: invertibility implies $\mathbb{E}[y_{jt} \mid \mathbf{o}_{jt}, \mathbf{o}_{jt-1}] = \mathbb{E}[y_{jt} \mid \mathbf{o}_{jt}]$. Estimate $y_{jt} = \mathbf{o}'_{jt-1}\beta + \psi(\mathbf{o}_{jt}) + e_{jt}$ and test $\beta = 0$. Rejected almost everywhere (Spain ESEE; US NBER-CES).
- Consequence: step 1 delivers planned output minus $\zeta_{jt} = \omega_{jt} - \mathbb{E}[\omega_{jt} \mid \mathbf{o}_{jt}]$. With an AR(1), the step-2 moment is off by $\rho\, \mathbb{E}[\zeta_{jt-1} \mid \mathbf{z}_{jt}]$, and k_{jt} is no longer a valid instrument (it was chosen knowing ω_{jt-1}).
- Fix: make the step-1 conditioning set contain the instruments, $\mathbf{o}_{jt-1} \supseteq \mathbf{z}_{jt}$, so add the lead k_{jt+1} to step 1. This is exact for an AR(1) and a first-order correction otherwise. Add every demand and conduct covariate you have; a Neyman-orthogonal step 2 lets step 1 use ML.

Monte Carlo (truth: mean log markup 0.25, $g'(0) = 0.70$)	Standard ACF	$+ k_{t+1}$	$+ k_{t+1}, p_t$
AR(1) productivity: mean log markup	-0.24	0.25	0.25
Nonlinear productivity: mean log markup	-0.15	0.30	0.25

Marginal Cost as a Sufficient Statistic (Jaumandreu 2025)

- With market power the FOC is $MR_{jt}\,\partial F/\partial X\, e^{\omega_{jt}} = w^X_{jt}$. Marginal revenue depends on conduct, rivals, and unobserved demand: it is a correspondence, not a function of observables. Proxies built on p_{jt} (including p in the ACF first stage) are inconsistent.
- But $MR = mc$, and mc is a single-valued function of (k, w, Y, ω): a sufficient statistic for conduct and demand. With no FOC wedges, $mc = AVC/\nu$.
- Cost-share proxy, robust to any conduct (needs g linear):

$$\omega_{jt-1} = y_{jt-1} - f(\mathbf{x}_{jt-1}) + \ln \frac{\nu_{jt-1}\,\alpha^X_{jt-1}/\alpha^{\mathcal{V}}_{jt-1}}{\theta^X_{jt-1}} - \varepsilon_{jt-1}.$$

- Two Hausman tests: market power (cost-share vs revenue-share proxy) and specification (dynamic panel vs cost-share proxy).
- Compustat 1960–2018 (5,621 firms): perfect competition is rejected; Cobb–Douglas fails the specification test (dynamic panel $\widehat{\beta}_K < 0$, $\hat{\nu} > 1.1$); a translog with labor-augmenting productivity passes, with $\hat{\nu} = 0.70$–0.85.

Testing the Production Approach (Raval 2023)

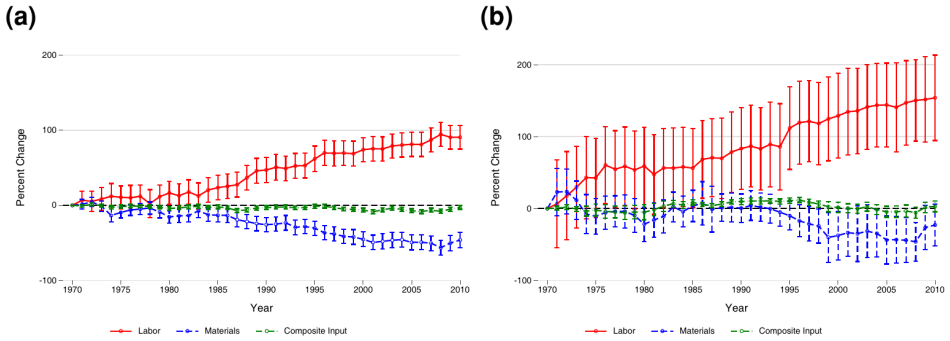
Any flexible input should give the same markup. ACF, translog, Hicks-neutral productivity, seven datasets:

	90/50 ratio of markups			Elasticity of μ^L
Dataset	Labor	Materials	Labor + materials	w.r.t. μ^M
Chile	2.03	1.39	1.17	-0.16
Colombia	1.82	1.43	1.17	-0.28
India	2.95	1.29	1.14	-0.53
Indonesia	3.12	1.46	1.16	-0.48
US (Compustat)	3.44	2.25	2.13	-0.50
Southern Europe	1.79	1.27	1.10	-0.32
US retailer (stores)	1.30	1.03	1.02	-10.08

Raval (2023), Tables 2–3; regressions include year and industry effects.

- Labor markups are far more dispersed, and labor and materials markups are negatively correlated everywhere.
- Not labor frictions: energy and raw-materials markups also disagree. Not local input prices or monopsony: the retailer result survives MSA-by-year effects.

Opposite Trends (Raval 2023, US Compustat)



Markup change relative to 1970, translog: (a) unweighted, (b) cost-weighted. Labor (red) +91% and +154%; materials (blue) -46% and -23%; labor + materials (green) +3%. Which input you choose decides whether US markups rose.

Why: Labor-Augmenting Productivity

CES with Hicks-neutral ω and labor-augmenting ω^L (Raval 2023, eqs. 8–10):

$$\frac{w^L_{jt} L_{jt}}{w^M_{jt} M_{jt}} = \left(\frac{w^L_{jt}}{w^M_{jt}}\right)^{1-\sigma} \left(\frac{\alpha_l}{\alpha_m}\right)^{\sigma} e^{(\sigma-1)\,\omega^L_{jt}},$$

where σ is the elasticity of substitution (not a choice probability).

- With $\sigma < 1$, higher ω^L lowers labor's cost share. A Hicks-neutral estimate attributes this to markups: $\hat{\mu}^L$ too high, $\hat{\mu}^M$ too low, hence the negative correlation.
- Cobb–Douglas makes θ^L/θ^M a constant, and translog a deterministic function of inputs. Neither has an error term to absorb variation in cost ratios.
- Evidence that $\sigma < 1$: Raval (2019, RAND) estimates a capital–labor $\sigma \approx 0.3$–0.5 from wage differences across 722 US commuting zones (amenity, Bartik, and BGS instruments). Labor-augmenting productivity is persistent and correlated with size, exports, and growth.
- Doraszelski and Jaumandreu (2019): DLW's exporter premium flips sign when computed from materials instead of labor (Spain; Slovenia).

Allowing for Heterogeneous Technology

- Demirer (2026): gross output with Hicks-neutral and labor-augmenting productivity; only homothetic separability of (L, M).
  - Invert the ratio of flexible-input demands for ω^L, then use an Imbens–Newey control variable for ω.
  - Functional dependence again: input–output variation identifies only the sum $\nu = \theta^L + \theta^M$ (exactly what the markup needs). The split comes from the FOCs, $\theta^L/\theta^M = \alpha^L/\alpha^M$, so labor and materials markups agree by construction.
  - Cobb–Douglas overstates aggregate markups by 8.5–16 pp across countries. US manufacturing: +6.6 pp (1.35 in the 1960s to 1.41 in the 2010s) vs +18 pp under Cobb–Douglas.
- Raval's flexible cost shares (2019 WP; Raval 2022): bin producers by $w^L L/(w^M M)$, a signal of ω^L, and use cost shares within bins. Labor and materials markups become positively correlated; markups rise 2–9% per doubling of sales.
- Doraszelski and Jaumandreu (2019): translog with both productivities, estimated jointly with the markup (dynamic panel or OP/LP). Exporters and non-exporters have about the same markups, and DLW-style markups are biased up.
- Raval's analogy: demand estimation learned to allow heterogeneity (random coefficients). The production approach needs heterogeneity in technology.

Every Wedge Looks Like a Markup

The production approach identifies θ^X/α^X: the markup times any other wedge in the FOC for X.

- Adjustment costs: $\theta^X/\alpha^X = \mu\,(1+\tau^X)$ (DLW fn. 50); $\mu\,(1+\kappa')$ (Bond et al. 2021).
- Input market power (Rubens 2023), with markdown $\psi^V = 1+$ inverse input-supply elasticity:

$$\mu = \left(\psi^L \frac{\alpha^L}{\theta^L} + \psi^M \alpha^M\right)^{-1},$$

  which nests $\mu = \theta^L/\alpha^L$ (DLW), $\mu = \theta^L/(\alpha^L\psi^L)$ (Morlacco), and $\mu = (\alpha^L/\theta^L + \alpha^M)^{-1}$ (Leontief, no markdowns).
- Production data alone cannot split a markup from a markdown. Rubens adds an input-supply model; the alternative is a downstream demand model.
- Rubens's application: a 2002 order closed small Chinese cigarette makers (340 $\rightarrow$ 167). Leaf's revenue share fell from 0.41 to 0.28 while labor's stayed near 0.03, consistent with rising oligopsony power over farmers.

What Each Approach Assumes

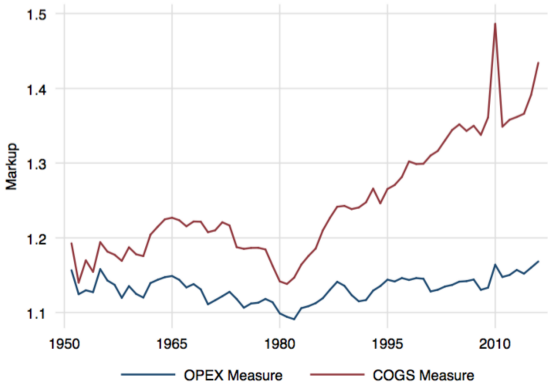
Method	ω_{t-1} replaced by	Rules out	Allows
Cost shares	—	adjustment costs; $\mu \neq 1$ (revenue shares) or non-CRS (cost shares)	any f, any demand (cost-share version)
Dynamic panel	lagged production function	nonlinear g; makes selection hard	unobserved cost and demand shocks; any conduct
OP	investment demand	capital-cost heterogeneity; wage and demand shocks	selection; nonparametric g
LP	materials demand	wage and demand shocks; first stage fails (functional dependence)	zero investment
ACF	conditional materials demand	unobserved w^M; demand shocks	dynamic labor; wage shocks
GNR	FOC (share regression)	$\mu \neq 1$ unless demand is modeled; wedges in m	gross output; nonparametric f
Doraszelski–Li	step 1 with k_{t+1}	very nonlinear g with large Var(ζ)	demand shocks; unobserved rivals
Jaumandreu	cost-share FOC ($mc = AVC/\nu$)	nonlinear g; FOC wedges	any conduct
Demirer	ratio of input demands + control variable	non-homothetic (L, M); demand shocks	labor-augmenting productivity

Aggregation and Misallocation

The Rise of Market Power?

- De Loecker, Eeckhout and Unger (2020): the sales-weighted average firm markup in Compustat is 1.34 in the 1960s, 1.21 in 1980 and 1.61 in 2016. The median is flat and the upper tail rises; about two-thirds of the rise is reallocation toward high-markup firms. DLW with COGS as the variable input, sector-year Cobb–Douglas.
- Every piece of this lecture bears on that number:
  - which costs: COGS alone, or COGS plus selling, general and administrative costs (Traina 2018; next slide);
  - which input: labor, materials, or COGS give opposite trends (Raval 2023);
  - which technology: Cobb–Douglas vs heterogeneous technology, +18 vs +6.6 pp in US manufacturing (Demirer 2026);
  - which identification: known returns to scale give cost-weighted Compustat markups of about 1.04–1.20 (Flynn–Gandhi–Traina 2019);
  - which data: revenue data cannot pin down levels, but trends can survive (Bond et al. 2021; De Ridder et al. 2026);
  - which average: Edmond–Midrigan–Xu (below).

Which Costs Are Variable? (Traina 2018; DLEU 2020)



Sales-weighted Compustat markups with COGS (red) or operating expenses, COGS + SG&A (blue), as the variable input. Traina (2018), Fig. 2.

- DLEU's method, with operating expenses (COGS + SG&A) as the variable input.
- SG&A rose from about 12% to 22% of operating expenses. A constant omitted share biases the level; a rising one biases the trend.
- COGS: 1.15 (1980) to about 1.40. Operating expenses: about 1.15 (1950), under 1.10 (early 1980s), back to the 1950 level. Reweighting to the Census size distribution makes it flat or falling.
- DLEU's reply: on their sample his measure still rises, 1.08 to 1.18 (1980–2016); with SG&A as an input the markup rises from about 1.0 to 1.3; the profit rate rises from 1% to 8% while overhead goes from 15% to 21% of total cost.

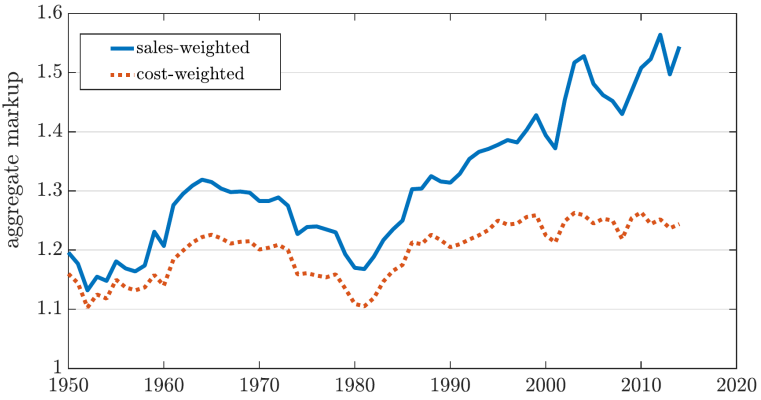
Which Average Markup? (Edmond, Midrigan and Xu 2023)

The markup that distorts aggregate input choices is the cost-weighted mean. With constant returns to the variable bundle, $\mu_{jt} = R_{jt}/VC_{jt}$; with sales shares $s_{jt} = R_{jt}/\sum_k R_{kt}$:

$$\mathcal{M}_t \equiv \sum_j \mu_{jt} \frac{VC_{jt}}{\sum_k VC_{kt}} = \frac{\sum_j R_{jt}}{\sum_j VC_{jt}} = \left(\sum_j \frac{s_{jt}}{\mu_{jt}} \right)^{-1} \leq \sum_j s_{jt}\, \mu_{jt}.$$

- $\mathcal{M}_t$ is aggregate revenue over aggregate variable cost: the wedge in the aggregate FOC. The steps use only $\theta^X = \mu\, \alpha^X$, so they hold for any demand system or conduct (their App. A).
- The inequality is harmonic $\leq$ arithmetic mean, with equality only if markups are equal. The sales-weighted mean overweights high-markup firms, so it overstates the aggregate distortion whenever markups are dispersed.
- Compustat 2012: 1.26 cost-weighted (1.21 counting SG&A as cost) vs about 1.6 sales-weighted.

Cost-Weighted vs Sales-Weighted Markups, Compustat



Edmond, Midrigan and Xu (2023), Figure 8 (NBER w24800 version). The sales-weighted average rises from about 1.17 in 1980 to about 1.55; the cost-weighted average is flat near 1.25 after 2000.

How Costly Are Markups? (Edmond, Midrigan and Xu 2023)

Kimball demand: larger, more productive firms face less elastic demand and charge higher markups.

- Benchmark aggregate markup 1.15 (Barkai 2017). Removing all markup distortions is worth 7.5% of consumption:
  - about 2/3 from the aggregate markup, which acts like a uniform output tax;
  - about 1/3 from misallocation (a TFP loss of 1.2%, against 8.4% under CES with the same aggregate markup);
  - almost nothing from inefficient entry.
- With the Compustat cost-weighted markup (1.25): 18.6%, four-fifths from the aggregate markup.
- Policy:
  - entry subsidies barely move $\mathcal{M}$, because direct and compositional effects offset;
  - size-dependent taxes lower $\mathcal{M}$ but worsen misallocation: the distorted economy has *too little* concentration.
- DLW markups on Taiwanese producers discipline the key demand parameter (how fast elasticities fall with size).

Misallocation: Hsieh and Klenow (2009)

- Benchmark: CES demand, CRS Cobb–Douglas technology, common input prices, no adjustment costs. Marginal revenue products are equal across producers, and $\text{TFPR}_{jt} = \ln \mu + \ln c(\mathbf{w}_t) - \ln \bar{p}_t$ is the same for every firm (shown earlier).
- So any TFPR dispersion is read as wedges. With an output wedge τ_{jt}^{Y} and a capital wedge τ_{jt}^{K}:

$$\text{TFPR}_{jt} = \text{const}_t + \beta_k \ln\left(1 + \tau_{jt}^{K}\right) - \ln\left(1 - \tau_{jt}^{Y}\right).$$

- Equalizing marginal products to the US level would raise manufacturing TFP by 30–50% in China and 40–60% in India.
- But outside the benchmark TFPR also varies with demand and markups (Haltiwanger, Kulick and Syverson 2018; FHS 2008), and with the timing of inputs (next slide).

...or Dynamics? (Asker, Collard-Wexler and De Loecker 2014)

Capital is chosen a period ahead. After optimizing out the flexible inputs, log revenue is $r_{jt} = c_t + \gamma\, k_{jt} + \tilde{\omega}_{jt}$, with $\gamma < 1$ and revenue productivity $\tilde{\omega}_{jt} = \rho\, \tilde{\omega}_{jt-1} + \xi_{jt}$. With no distortions the firm sets $\mathbb{E}_{t-1}[\text{MRPK}_{jt}] = w_t^K$, so

$$\ln \text{MRPK}_{jt} = \ln \gamma + c_t - (1-\gamma)\, k_{jt} + \tilde{\omega}_{jt} \quad \Longrightarrow \quad \ln \text{MRPK}_{jt} = \ln w_t^K - \ln \mathbb{E}\left[e^{\xi}\right] + \xi_{jt}.$$

- MRPK is dispersed with no distortion: its cross-sectional variance is σ_ξ^2, the volatility of revenue productivity, which ACDL measure from a revenue production function. Adjustment costs add more.
- Measured volatility and MRPK dispersion line up across industries in 8 countries and across 33 countries in the World Bank Enterprise Surveys.
- A dynamic investment model with adjustment costs fits the cross-industry and cross-country dispersion with an uncentered R^2 of about 0.8–0.9.
- Lesson: static marginal-product gaps are weak evidence of policy distortions. The production function, the timing of inputs, and the demand system all matter.

Takeaways

- Be explicit about timing, the scalar unobservable, and what moves a flexible input besides ω. That last item is your exclusion restriction.
- Use ACF-style conditional proxies rather than LP's first stage. For gross output use GNR (price takers) or a Leontief argument, and cross-check against dynamic panel.
- Under market power, put demand and conduct shifters (and k_{t+1}) into step 1, or use cost-share forms of the FOC.
- Put firm actions (exporting, R&D) into the law of motion g, not into a second-stage regression on $\widehat{\omega}$.
- Markups:
  - levels need output prices; dispersion and trends survive revenue data if technology is common;
  - check that different inputs agree, and allow labor-augmenting technology when they do not;
  - every wedge (adjustment costs, markdowns) looks like a markup;
  - aggregate with cost weights (harmonic sales weights) for welfare.
- The production approach complements the demand approach. It measures market power across many industries; it cannot simulate a merger.